

Software Project Management Document

Qdreon | Project Charter

CPE 2201 SOFTWARE DESIGN

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1.0	04/21/2026	Connecting Docs to gsheets
2.0	04/26/2026	Created the Project Definition, and Project Organization(Subject to changes)
3.0	05/04/2026	Finalized Project Plan, Financials, Risks, and Considerations based on SWDD and SRS.
4.0	05/05/2026	Updated Responsibilities for the roles, recalculated Resource Effort via version history analytics, and integrate 16-week Gantt Chart.

Document Approvals

Role	Name	Signature	Date
Project Sponsor	LT3 (Stakeholders)		05/04/26
Project Review Group	Godwin S. Montserrat, PhD & Engr. Imperial		
Project Manager	Kisha Belocura		05/04/26
Quality Manager	Mickel Angelo Castroverde		05/04/26
Procurement Manager	Dale Ryoko Jose V. Cantago		05/04/26
Communications Manager	Reuel Matthew Arcilla		05/04/26
Project Office Manager	Cyrus Alain Durango		05/04/26

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1 Executive Summary

Company	Qdreon
Project Name	Gordon Ramsay Restaurant Reservatio
Proposed Start Date	February 02, 2026
Proposed End Date	May 24, 2026
Preparer	Kisha Faye Belocura
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Summary of the Project:

This software is a web-based, single-tenant application replacing manual reservation logbooks with scalable digital platform featuring a Customer Portal and an Admin Dashboard with real-time interactive floor plan management.

Executed by Lab Team 1 (Qdreon) utilizing Agile and COMET methodologies over a 16-week timeline, utilizing Supabase as a Backend-as-a-Service (BaaS).

Primary risks and issues include real-time database desynchronization and double-booking concurrency conflicts. These are mitigated via Supabase PostgreSQL row-locking and strict "Offline Mode" system health monitoring.

The assumptions and constraints of this project is that we assume that the system is connected to a constant high-speed, stable, internet connection at the restaurant and basic digital literacy of staff. Constrained by a 3-second UI load limit, a 1-second lock resolution, and simulated-only payment processing to avoid PCI-DSS liability during the MVP phase.

2 Project Definition

This section describes what the project sets out to achieve. It outlines the vision, objectives, scope, and deliverables based on the Gordon Ramsay Restaurant Reservation System requirements and design.

2.1 Vision

The vision of this project is:

To deliver a robust, scalable, and legally compliant restaurant reservation system that replaces manual logbooks with a centralized, real-time digital platform, ensuring frictionless customer booking and efficient restaurant operations.

2.2 Objectives

Business Objectives

- To streamline reservation handling by eliminating manual logbooks and reducing booking errors.
- To enhance customer satisfaction through real-time availability, automated notifications, and personalized CRM features.
- To reduce financial losses from no-shows by enforcing deposit handling and automated cancellation workflows.

- To improve operational efficiency by providing administrators with a static, interactive floor plan grid and system health monitoring.

Technology Objectives

- To implement a Supabase-based backend ensuring ACID-compliant PostgreSQL transactions and real-time synchronization.
- To enforce secure authentication and role-based access control (RBAC) for customers and administrators.
- To integrate external services (SMTP Email, simulated Payment Gateway) for notifications and deposits without storing sensitive financial data.
- To ensure compliance with Republic Act No. 10173 (Data Privacy Act of 2012) by implementing account deletion and consent mechanisms.

2.3 Scope

The scope of the project includes:

- **Processes to change:** Reservation booking, waitlist handling, deposit management, and guest CRM tracking.
- **Organizational areas affected:** Front-desk operations, customer service, and restaurant administration.
- **Locations impacted:** Gordon Ramsay Restaurant (single dedicated site).
- **Data altered:** Reservation records, customer profiles, CRM notes, menu items, and waitlist queues.
- **Applications installed/altered:** Customer Portal (web-based), Admin Dashboard (management interface).
- **Technologies deployed:** Supabase BaaS, PostgreSQL database, SMTP Email integration, simulated Payment Gateway.

Excluded from scope: *Multi-tenant franchise management, SMS notifications, and physical POS hardware integration.*

2.4 Deliverables

Item	Components	Description
Customer Portal	Account Management, Search & Booking, Waitlist	Web interface for diners to register, view menus, book tables, join waitlists, and manage reservations.
Admin Dashboard	Floor Plan Grid, Reservation Manager, CRM, Menu Management	Operational interface for staff to monitor table statuses, manage bookings, track guest history, and update menus.
Backend Services	Supabase Auth, PostgreSQL, Real-Time Subscriptions	Core logic and data storage ensuring secure authentication, concurrency control, and synchronization.
External Integrations	SMTP Email, Simulated Payment Gateway	Automated notifications, calendar invites, and deposit handling without storing sensitive financial data.
Compliance Features	Account Deletion, Consent Mechanisms	Legal compliance with Data Privacy Act of 2012 and simulated PCI-DSS standards.

3 Project Organisation

This section identifies the customers, stakeholders, roles, responsibilities, and reporting structure for the Gordon Ramsay Restaurant Reservation System project.

3.1 Customers

Customer	Representative
Diners (restaurant guests)	Customer Portal users
Restaurant Administrators	Front-desk staff and managers
Gordon Ramsay Restaurant Management	Executive sponsor for operational oversight

3.2 Stakeholders

Stakeholder	Interested in
CEO / Restaurant Owner	Alignment with business vision and customer satisfaction
Financial Controller	Budget compliance and deposit handling
Health & Safety Officer	Compliance with operational standards
Government / Regulatory Bodies	Data Privacy Act compliance (RA 10173)
Industry Body	Adherence to best practices in hospitality technology

3.3 Roles

Role	Organisation	Resource Name	Assignment Status	Assignment Date
Project Sponsor	Gordon Ramsay Restaurant	Restaurant Owner (LT3)	Assigned	Feb 02, 2026
Project Review Group	USC Faculty / LT3 Stakeholders	Professors & Stakeholder Team	Assigned	Feb 02, 2026
Project Manager	Lab Team 1	Kisha Faye Belocura	Assigned	Feb 02, 2026
Quality Manager / Lead Developer	Lab Team 1	Mickel Angelo Castroverde	Assigned	Feb 02, 2026
Communications Manager / UI/UX	Lab Team 1	Cyrus Alain Durango	Assigned	April 2026

Procurement Manager / Backend Dev	Lab Team 1	Dale Ryoko Jose V. Cantago	Assigned	April 2026
Project Office Manager / Backend Dev	Lab Team 1	Reuel Matthew Arcilla	Assigned	April 2026

3.4 Responsibilities

List the generic responsibilities for each role identified. A full list of the responsibilities, performance criteria and skills required should be documented within a separate *Job Description* for each project role.

Project Manager

- Documents and maintains detailed Project Plan, Resource Plan, and Quality Plan.
- Assigns and manages resources according to the defined 16-week MVP scope.
- Implements project processes: time, cost, quality, change, risk, issue, and acceptance management.
- Monitors and reports project performance and ensures compliance with academic IEEE standards.
- Calculates and tracks Expected Monetary Value (EVM) and team % effort metrics.

Quality Manager and Lead Developer

- Leads core backend architecture design, including Supabase PostgreSQL schema and row-locking logic.
- Defines and enforces quality assurance processes across all development phases.
- Validates the system meets strict performance metrics (UI latency <3s, concurrency resolution <1s).
- Reports quality findings and bug tracking via Defects Log to the Project Manager.
- Oversees the execution of formal Test Scripts (TC-1.1 – TC-6.2).

Communications Manager and UI/UX Developer

- Designs and develops the React/[Next.js](#) fronted Customer Portal and Admin Dashboard.
- Builds the real-time static interactive floor plan and ensures visual color-coding accuracy.
- Manages communication flow between developers, stakeholder (LT3), and faculty.
- Prepares project updates, meeting minutes, and facilitates stakeholder feedback loops.

Procurement Manager and Backend Developer

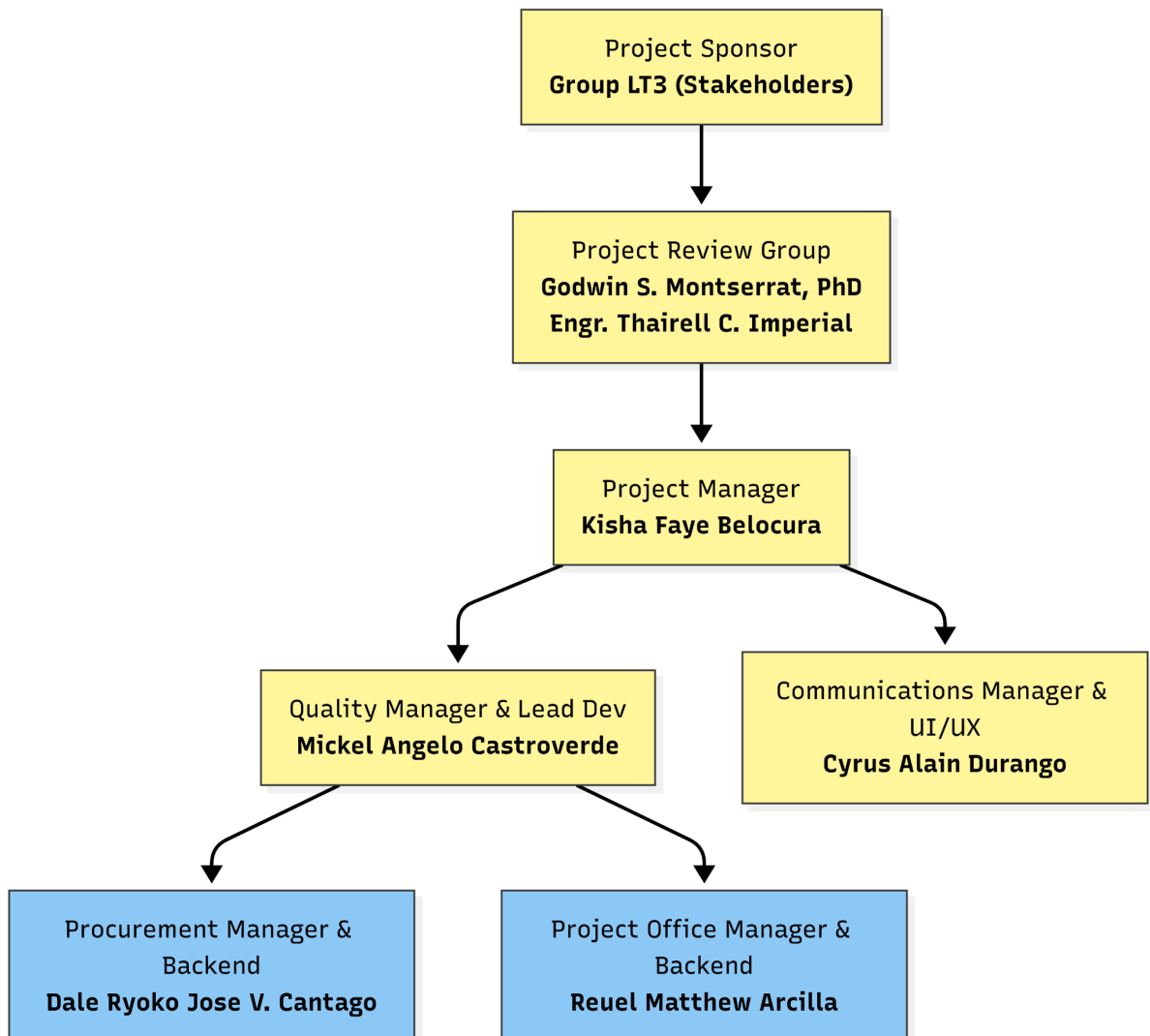
- Oversees the acquisition and integration of external API services (SMTP Email and simulated Payment Gateways).
- Assists in developing backend modules, specifically automated notification payloads and waitlist triggers.
- Ensures external integrations comply with procurement policies, budget constraints (Github Student Pack), and simulated PCI-DSS

Project Office Manager and Backend Developer

- Manages configuration and document version control via Google Workspace/Github.
- Develops backend Supabase Role-Based Access Control (RBAC) authentication and Guest CRM Tables.
- Ensures that legal compliance requirements (Data Privacy Act 2012 / Right to Erasure) are properly implemented in the database architecture.

3.5 Structure

Organisation Chart



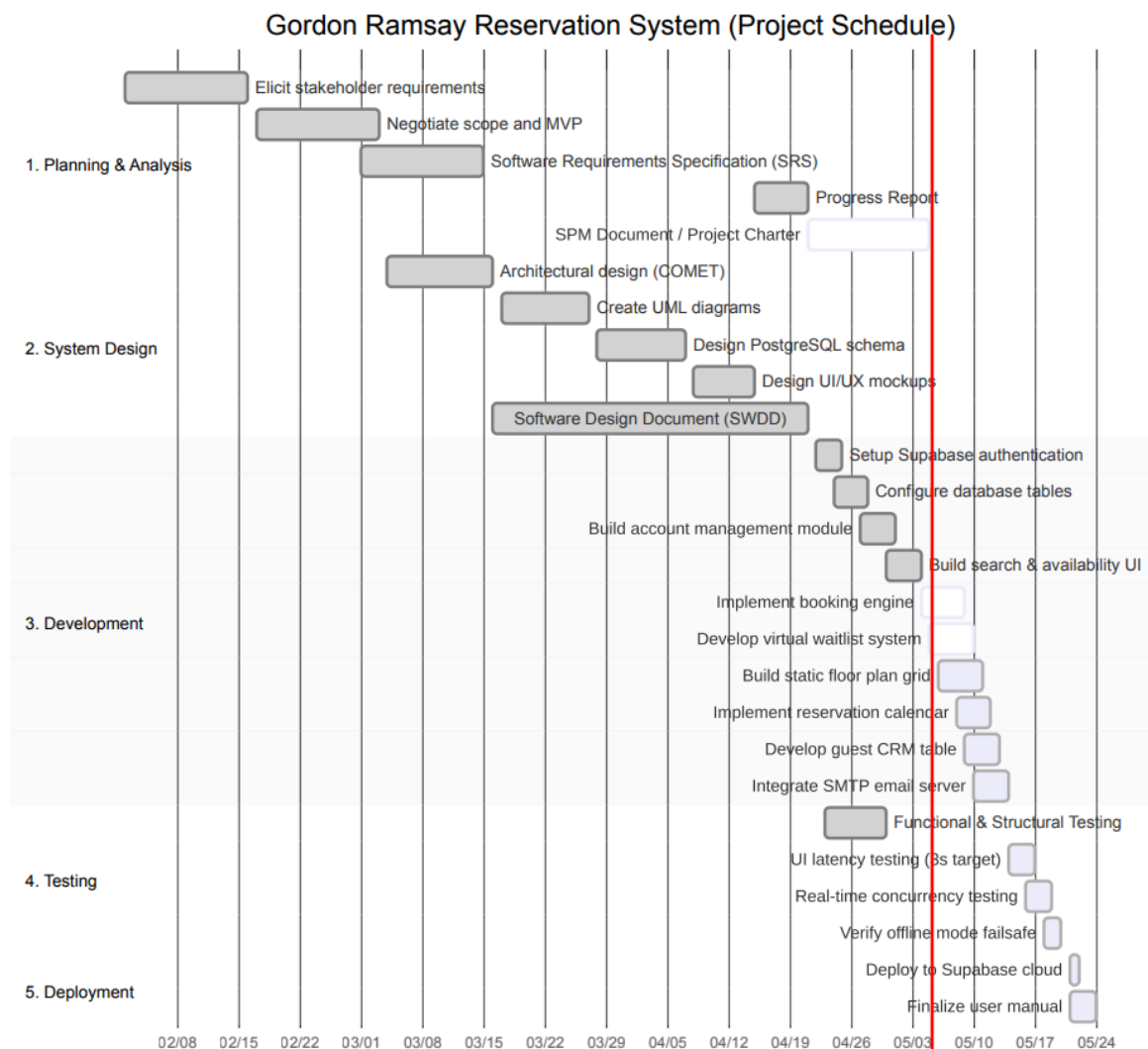
4 Project Plan

4.1 Approach

Describe the approach to be taken to implement each of the phases within the project.

Phase	Approach
Initiation	Elicit stakeholder requirements (LT3), negotiate scope limitations (MVP), and establish the initial project vision and business case.
Planning	Utilize Agile methodologies to create User Stories, estimate Epic points, and map the architecture using COMET (UML, DFDs, MVS modeling). Develop WBS and Gantt Chart
Execution	Execute 4 development sprints: (1) Environment/Auth Setup, (2) Core Booking & Concurrency Engine, (3) External API Integrations, and (4) QA Testing
Closure	Perform system latency tests, execute the formal Test Plan (TC-1.1 to 6.2), resolve bugs via the Defects Log, finalize deployment to Supabase cloud, and deliver User Manuals.

4.2 Overall Plan



Milestones

Milestone	Date	Description
SRS & SWDD Finalized	04/21/2026	Formalization of requirements, constraints, and architecture blueprints.
Sprint 1: Backend Setup	Week 1 Dev	Initialization of React/ Next.js frontend, Supabase database migrations, and RBAC authentication.
Sprint 2: Booking Engine	Week 2 Dev	Completion of the Customer Portal, PostgreSQL row-locking, and the real-time static interactive floor plan.
Sprint 3: API Integration	Week 3 Dev	Successful connection of SMTP Email for automated waitlist/ICS invites and simulated Payment.
QA & Deployment	Week 4 Dev	Execution of Pass 1 Test Scripts, bug resolution, and final production deployment

Dependencies

Project Activity	Impacts on	Impacted on by	Criticality	Date
Supabase Environment Setup	Core Booking Engine & Waitlist Logic	Approval of Data Dictionary / Class Model	High	Week 1
Row-Locking Logic Integration	Double-Booking Prevention	Supabase Database Setup	High	Week 2
Payment Gateway Integration	Deposit Status & Reservation Confirmation	Core Booking Engine Completion	Medium	Week 3

4.3 Resource Plan

Role	Start Date	End Date	% Effort
Project Manager	02/02/2026	05/24/2026	20%
Lead Developer	02/02/2026	05/24/2026	25%
UI/UX Developer	02/02/2026	05/24/2026	20%
Backend Developers	02/02/2026	05/24/2026	35%

4.4 Financial Plan

Note: As an academic Minimum Viable Product (MVP), actual cash expenditure is ₱0 or 0\$. The project relies entirely on the Github Student Developer Pack and open-source free tiers. The “Value” column represents the enterprise equivalent cost subsidized by these academic licenses.

Category	Cost	Value
Cloud Hosting (Frontend)	DigitalOcean / Microsoft Azure (via GitHub Student Pack)	\$200.00 platform credit
Custom Domain & SSL	Name.com / Namecheap (via GitHub Student Pack)	~\$25.00 / year waived
Backend as a Service (BaaS)	Supabase (Free Tier)	~\$25.00 / month waived
System Monitoring & QA	Sentry / New Relic (via GitHub Student Pack)	~\$300.00 / month waived
Developer Tools & CI/CD	GitHub Pro & GitHub Copilot (via GitHub Student Pack)	~\$30.00 / user/month waived
Automated Email Testing	Testmail API (via GitHub Student Pack)	~\$20.00 / month waived
Development Labor	Lab Team 1 (5 Developers over 16-weeks)	Academic / \$0 actual

4.5 Quality Plan

Process	Description
Quality Management	Ensuring UI load times stays under 3 seconds and concurrency locks resolve within 1 second.
Change Management	Strict adherence to the negotiated MVP scope; any new feature requires stakeholder renegotiation
Risk & Issue Management	Utilizing Expected Monetary Value (EMV) estimations to mitigate scope creep and timeline delays.
Document Management	Maintaining version control for the SRS, SWDD, and Test Execution Plans via Google Workspace/Github.
Acceptance Management	Passing Test Scripts TC-1.1 through TC-6.2 to verify FR-1 through FR-13

5 Project Considerations

5.1 Risks

Description	Likelihood	Impact	Mitigating Actions
Simultaneous Booking Conflict (Two diners select the exact same table)	Medium	High	Utilize Supabase's 1-second PostgreSQL row-locking to abort secondary transactions and prevent database overlaps.
Internet Outage at Restaurant (Data Desynchronization)	Low	High	Implement "System Health Monitoring" (FR-13). Display "Offline Warning" and disable table edits until connection is restored.
PCI-DSS Compliance Violation (Data Breach of customer cards)	Low	High	Strictly forbid raw PAN storage (LEG-2); utilize tokenized simulated checkout gateways only.
Data Loss / Server Crash	Low	Critical	Rely on Supabase automated Point-in-Time Recovery (PITR) and daily backups (SAF-1).

To complete this section thoroughly, it may be necessary to undertake a formal Risk Assessment (by documenting a *Risk Management Plan*). To reduce the likelihood and impact of each risk's eventuating, clear 'mitigating actions' should be defined.

5.2 Issues

Summarize the highest priority issues associated with the project. Issues are defined as "any event which currently adversely affects the ability of the solution to produce the required deliverables". Complete the following table:

Description	Priority	Resolution Actions
Scope Creep (Multi-tenant system request)	High	Successfully negotiated down to a single-tenant architecture during Stakeholder Meeting 2 to ensure timely MVP delivery.
Waitlist Automation Logic Conflicts	High	Resolved by adding a 10-minute acceptance window constraint and a 60-minute pre-closing cutoff in the SRS (FR-5).

5.3 Assumptions

- The restaurant is assumed to have a high-speed, stable, and constant internet connection to maintain real-time double-booking prevention.
- The restaurant management will provide standard touchscreen terminals (e.g., iPads, Android tablets) for the host stand.
- The restaurant must use this system as its exclusive reservation tool to ensure static grid is accurate.
- Storage of customer data will be compliant with the Data Privacy Act of 2012 in the Philippines, assuming users consent upon registration.

5.4 Constraints

Methodology

System architecture and documentation must strictly follow COMET (Concurrent Object Modelling and Architectural Design Method) and UML standards.

Performance Time

The application must load within 3 seconds; lock resolution must occur within 1 second.

Security Limitations

The project is strictly prohibited from capturing real credit card Primary Account Numbers (PAN) to mitigate legal liability during this phase.

Schedule

The project is bound by a strict 16-week academic delivery timeline.

6 Appendix

6.1 Supporting Documentation

Gordon Ramsay Reservation System - Software Requirements Specification (SRS v1.4)

Gordon Ramsay Reservation System - Software Design Document (SWDD-QDR-01)

Qdreon Progress Report (April 21, 2026)

Requirement Traceability Matrix & Data Dictionary (Referenced in SWDD Appendix A)

Group LT3 Stakeholder Meeting Logs (Recorded Sessions 1 & 2)